```
!pip install uv
!uv pip install pandas matplotlib numpy scipy scikit-learn
```

```
Collecting uv
  Downloading uv-0.10.8-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (11 kB)
Downloading uv-0.10.8-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (23.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 23.3/23.3 MB 38.8 MB/s eta 0:00:00
Installing collected packages: uv
Successfully installed uv-0.10.8
Using Python 3.12.12 environment at: /usr
Audited 5 packages in 130ms
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler, PolynomialFeatures
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.linear_model import Ridge
from urllib.request import urlretrieve

# Set random seed for reproducibility
np.random.seed(69) # nice
import os
```

```
def load_auto_mpg(sample_size=100, random_state=69):
    """
    Load Auto MPG dataset with specified sample size.
    Parameters:
    -----------
    sample_size : int or None
        Number of samples to randomly select from dataset.
        If None, use the full cleaned dataset.
    random_state : int
        Seed used for deterministic sampling.
    Returns:
    --------
    X : numpy array (n, 2) - features [weight, horsepower]
    y : numpy array (n, 1) - target [mpg]
    """
    # Load Auto MPG dataset (download only if missing locally)
    url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/auto-mpg.data'
    if not os.path.exists('auto-mpg.data'):
        urlretrieve(url, 'auto-mpg.data')

    column_names = ['mpg', 'cylinders', 'displacement', 'horsepower', 'weight',
                    'acceleration', 'model_year', 'origin', 'car_name']

    data = pd.read_csv('auto-mpg.data',
                   names=column_names,
                   sep=r'\s+',
                   na_values='?')
    data = data.dropna()

    # Sample specified number of records deterministically
    if sample_size is not None:
        sample_size = min(sample_size, len(data))
        data = data.sample(n=sample_size, random_state=random_state)

    # Extract features and target
    X = data[['weight', 'horsepower']].values
    y = data['mpg'].values.reshape(-1, 1)

    print(f"Dataset: {len(y)} cars")
    print(f"X shape: {X.shape}, y shape: {y.shape}")

    return X, y

def minMaxNormalize(T):
    T_norm = (T - np.min(T)) / (np.max(T)-np.min(T))
    return T_norm

X, y = load_auto_mpg(sample_size=None)

df_normalized = minMaxNormalize(X)

print(df_normalized)
```

```
[4.59167648e-01 7.45975658e-02]
```

```
 [4.59167648e-01 7.45975658e-03]
 [3.35492737e-01 2.35571260e-03]
 [3.59049863e-01 3.53356890e-03]
 [3.36474283e-01 2.74833137e-03]
 [3.96348645e-01 4.12249706e-03]
 [3.78680801e-01 3.72987829e-03]
 [3.93404005e-01 3.14095014e-03]
 [3.80643895e-01 4.31880644e-03]
 [4.25795053e-01 3.33725952e-03]
 [3.92422458e-01 3.72987829e-03]
 [4.58186101e-01 3.72987829e-03]
 [4.20887318e-01 5.49666274e-03]
 [4.24813506e-01 5.69297212e-03]
 [4.52296820e-01 5.69297212e-03]
 [5.04318806e-01 1.06007067e-02]
 [5.08244994e-01 5.49666274e-03]
 [6.25049077e-01 6.67451904e-03]
 [6.11307420e-01 5.88928151e-03]
 [5.60266981e-01 1.37416569e-02]
 [5.66156262e-01 1.45268944e-02]
 [6.61366313e-01 1.25638005e-02]
 [7.22222222e-01 1.15822536e-02]
 [5.91676482e-01 8.24499411e-03]
 [6.71181782e-01 7.65606596e-03]
 [5.02355713e-01 8.24499411e-03]
 [5.09226541e-01 8.24499411e-03]
 [4.61130742e-01 8.24499411e-03]
 [4.96466431e-01 7.65606596e-03]
 [4.86650962e-01 7.45975658e-03]
 [5.27875932e-01 8.63761288e-03]
 [5.53396152e-01 9.03023165e-03]
 [3.79662348e-01 5.49666274e-03]
 [3.88496270e-01 4.31880644e-03]
 [3.77699254e-01 4.31880644e-03]
 [4.08127208e-01 3.33725952e-03]
 [4.08127208e-01 4.71142521e-03]
 [4.14998037e-01 8.24499411e-03]
 [4.23831959e-01 5.69297212e-03]
 [4.31684335e-01 4.71142521e-03]
 [3.76717707e-01 4.12249706e-03]
 [3.76717707e-01 4.12249706e-03]
 [3.82606989e-01 4.12249706e-03]
 [5.69100903e-01 1.25638005e-02]
 [5.82842560e-01 7.65606596e-03]
 [4.98429525e-01 9.03023165e-03]
 [5.47506871e-01 1.29564193e-02]
 [5.14134276e-01 9.81546918e-03]
 [4.56223007e-01 7.45975658e-03]
 [5.70082450e-01 8.63761288e-03]
 [5.38672949e-01 7.85237534e-03]
 [4.09108755e-01 1.17785630e-03]
 [4.41499804e-01 7.45975658e-03]
 [5.06281900e-01 6.47820966e-03]
 [5.24931292e-01 7.06713781e-03]]
```

```
from sklearn.model_selection import train_test_split

def initial_split(X, y, test_size=0.2, random_seed=42):
    """
    Split data into development and test sets.
    Parameters:
    -----------
    X : numpy array (n, p)
    y : numpy array (n, 1)
    test_size : float (default 0.2)
    Proportion of data for test set
    random_seed : int (default 42)
    Random seed for reproducibility
    Returns:
    --------
    X_dev, y_dev : Development set (for training and CV)
    X_test, y_test : Test set (for final evaluation only)
    """
    X_dev, X_test, y_dev, y_test = train_test_split(
        X, y, test_size=test_size, random_state=random_seed
    )
    return X_dev, y_dev, X_test, y_test
```

```
from sklearn.preprocessing import PolynomialFeatures

def create_polynomial_features(X, degree):
    """
    Generate polynomial features up to specified degree.
    Parameters:
    -----------
    X : numpy array (n, p) - normalized features
    degree : int - polynomial degree
    Returns:
    --------
    X_poly : numpy array (n, m) - polynomial features (no bias column)
    Note: include_bias=False means no column of 1s is added
    """
```

```
    poly = PolynomialFeatures(degree=degree, include_bias=False)
    return poly.fit_transform(X)
```

```
X_simple = np.array([[0.5, 0.3], [0.7, 0.6], [0.2, 0.9]])
X_simple_poly2 = create_polynomial_features(X_simple, 2)
X_simple_poly3 = create_polynomial_features(X_simple, 3)

print("Baseline: ")
print(X_simple)
print("Polynomial 2nd: ")
print(X_simple_poly2)
print("Polynomial 3rd: ")
print(X_simple_poly3)
```

```
Baseline:
[[0.5 0.3]
 [0.7 0.6]
 [0.2 0.9]]
Polynomial 2nd:
[[0.5  0.3  0.25 0.15 0.09]
 [0.7  0.6  0.49 0.42 0.36]
 [0.2  0.9  0.04 0.18 0.81]]
Polynomial 3rd:
[[0.5   0.3   0.25  0.15  0.09  0.125 0.075 0.045 0.027]
 [0.7   0.6   0.49  0.42  0.36  0.343 0.294 0.252 0.216]
 [0.2   0.9   0.04  0.18  0.81  0.008 0.036 0.162 0.729]]
```

```
def get_fold(X, y, fold_number, total_folds):
    """
    Retrieve a specific fold for k-fold cross-validation.
    Parameters:
    -----------
    X : numpy array (n, p)
    y : numpy array (n, 1)
    fold_number : int (0-indexed: 0, 1, ..., total_folds-1)
    total_folds : int
    Returns:
    --------
    X_train, y_train : All samples except fold_number
    X_val, y_val : Only samples in fold_number
    """
    # Split
    X_split = np.array_split(X, total_folds)
    y_split = np.array_split(y, total_folds)

    # Remove consumed
    X_val = X_split.pop(fold_number)
    y_val = y_split.pop(fold_number)

    # Merge non-consumed
    X_train = np.concatenate(X_split, axis=0)
    y_train = np.concatenate(y_split, axis=0)

    return X_train, y_train, X_val, y_val
```

```
X_test = np.arange(10).reshape(-1, 1)
y_test = np.arange(10).reshape(-1, 1)
for fold in range(3):
    X_train, y_train, X_val, y_val = get_fold(X_test, y_test, fold, 3)
    print(f"Fold {fold}: Val={y_val.flatten()}, Train={y_train.flatten()}")
```

```
Fold 0: Val=[0 1 2 3], Train=[4 5 6 7 8 9]
Fold 1: Val=[4 5 6], Train=[0 1 2 3 7 8 9]
Fold 2: Val=[7 8 9], Train=[0 1 2 3 4 5 6]
```

Problem 3.1: Understanding Regularized Gradients (15 points)

Part A: Verify Dimensions (3 pts)

1. What is the shape of e^T?

$(1, n)$

2. What is the shape of $e^T X$?

$(1, m)$

3. What is the shape of w?

$(1, m)$

4. Does the gradient have the correct shape?

Yes. In $rac\partial L\partial w = -rac2ne^T X + 2\lambda w$, each term has shape $(1, m)$, so the gradient is $(1, m)$.

Problem 3.2: Implement Regularized Gradient Descent (10 points)

Modify your VectorizedMLR from Assignment 4 to include L2 regularization.

Key changes:

1. Add `lambda_reg` parameter
2. Update gradient for w
3. Update loss

Testing:

- Verify that `lambda_reg=0` reproduces unregularized results
- Verify that `lambda_reg=0.001` converges and has smaller weights than the unregularized results
- Show that losses decrease over iterations
- *In your PDF, highlight the three modified lines with comments*

```
import numpy as np

class VectorizedMLR:
    """
    Multiple Linear Regression: ŷ = X@wᵀ + b

    Works for ANY number of features:
        - p = 1: Simple Linear Regression
        - p > 1: Multiple Linear Regression

    REQUIRED Dimensions:
        X: (n, p) - n observations, p features
        w: (1, p) - 1 row, p features (ROW VECTOR)
        b: scalar
        ŷ: (n,1) - n predictions
    """

    def __init__(self, learning_rate=0.01, num_iterations=1000, lambda_reg=0.01):
        self.learning_rate = learning_rate
        self.num_iterations = num_iterations
        self.w = None  # Will be (1, p)
        self.b = 0.0
        self.loss_history = []
        self.lambda_reg = lambda_reg # Initialize Lambda Rate ------------- Changed

    def predict(self, X):
        """
        Matrix-based prediction: ŷ = X@wᵀ + b

        Input:
            X: numpy array (n, p)
        Returns:
            y_pred: numpy array (n,1) - predictions
        """
        y_pred = X @ self.w.T + self.b
        return y_pred

    def compute_loss(self, X, y):
        """Mean Squared Error: (1/n) * sum((y_pred - y)^2)"""
        n = X.shape[0]
        y_pred = self.predict(X)
        loss = (1 / n) * np.sum((y_pred - y) ** 2) + self.lambda_reg * np.sum(self.w ** 2) # Added Lambda Regularization Term
        return loss

    def compute_gradients(self, X, y):
        """
        Computes gradients for w and b.
        grad_w = dJ/dw
        grad_b = dJ/db
        """
        n = X.shape[0]
        y_pred = self.predict(X)
        error = y_pred - y # Shape (n, 1)

        # Gradient w:
        # Shape must be (1, p) to match self.w
        # Formula: (2/n) * error.T @ X
        # (1, n) @ (n, p) -> (1, p)
        grad_w = (2 / n) * (error.T @ X) + 2 * self.lambda_reg * self.w # ------ Added Lambda Regularization Term

        # Gradient b:
        # Scalar mean of errors * 2
        grad_b = (2 / n) * np.sum(error)

        return grad_w, grad_b

    def fit(self, X, y):
        """
        Apply gradient descent to optimize the parameters of the model.

        Input:
            X: numpy array (n, p)
            y: numpy array (n, 1)
```

```
        Returns:
            self for chained calls.
        """
        n_samples, n_features = X.shape
        self.w = np.zeros((1, n_features))
        self.b = 0.0
        self.loss_history = []

        # Gradient Descent Loop
        for i in range(self.num_iterations):
            # Calculate gradients
            grad_w, grad_b = self.compute_gradients(X, y)

            # Update parameters
            self.w = self.w - (self.learning_rate * grad_w)
            self.b = self.b - (self.learning_rate * grad_b)

            # Track loss
            current_loss = self.compute_loss(X, y)
            self.loss_history.append(current_loss)

        return self
```

Problem 3.3: Build and Test Baseline Model (10 points)

Report:

Metric	Development Set	Test Set
MSE	18.3410	18.3461
R^2	0.6974	0.6998

Analysis (2-3 sentences):

Based on the results, we're seeing very good performance with no overfitting. Shockingly, we can see that there is *better* performance in the "Test Set" than the "Development Set".

```
# Data loaded in earlier cell
X, y = load_auto_mpg()

# Split the data between train and test sets
X_train, y_train, X_test, y_test = initial_split(X, y, test_size=0.2, random_seed=42)
```

```
Dataset: 100 cars
X shape: (100, 2), y shape: (100, 1)
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

# Fit on train set and transform train set
X_train_normalized = scaler.fit_transform(X_train)

# only transform the test set
X_test_normalized = scaler.transform(X_test)
```

```
# Initialize model with no lambda regularization
model = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=0)

# Train the model on the normalized development data
model.fit(X_train_normalized, y_train)
```

```
<__main__.VectorizedMLR at 0x7b3f96bc5e20>
```

```
from sklearn.metrics import mean_squared_error, r2_score

# Predict on Development Set
y_dev_pred = model.predict(X_train_normalized)
dev_mse = mean_squared_error(y_train, y_dev_pred)
dev_r2 = r2_score(y_train, y_dev_pred)

# Predict on Test Set
y_test_pred = model.predict(X_test_normalized)
test_mse = mean_squared_error(y_test, y_test_pred)
test_r2 = r2_score(y_test, y_test_pred)

print(f"Dev MSE: {dev_mse:.4f}, Dev R2: {dev_r2:.4f}")
print(f"Test MSE: {test_mse:.4f}, Test R2: {test_r2:.4f}")
```

```
Dev MSE: 14.1843, Dev R2: 0.7448
Test MSE: 10.8202, Test R2: 0.7146
```

Problem 3.4: Cross-Validation for Different Configurations (30 points)

Part A (n=100, degree=10): Optimal $\lambda = 0$, Training MSE $= 20.4853$, Validation MSE $= 20.8997$.

Part B (n=100, degree=15): Optimal $\lambda = 0.01$, Training MSE $= 21.2814$, Validation MSE $= 22.01299$.

Part C (n=200, degree=10): Optimal $\lambda = 0$, Training MSE $= 20.4853$, Validation MSE $= 20.8997$.

```
# Load data and split (sample_size=100, degree=10)
X, y = load_auto_mpg(sample_size=100)
X_dev, y_dev, X_test, y_test = initial_split(X, y, test_size=0.2, random_seed=42)

lambdas = [0, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100]
degree = 10
total_folds = 5

# Arrays to store average MSE for each lambda
train_mse_averages = []
val_mse_averages = []
```

```
Dataset: 100 cars
X shape: (100, 2), y shape: (100, 1)
```

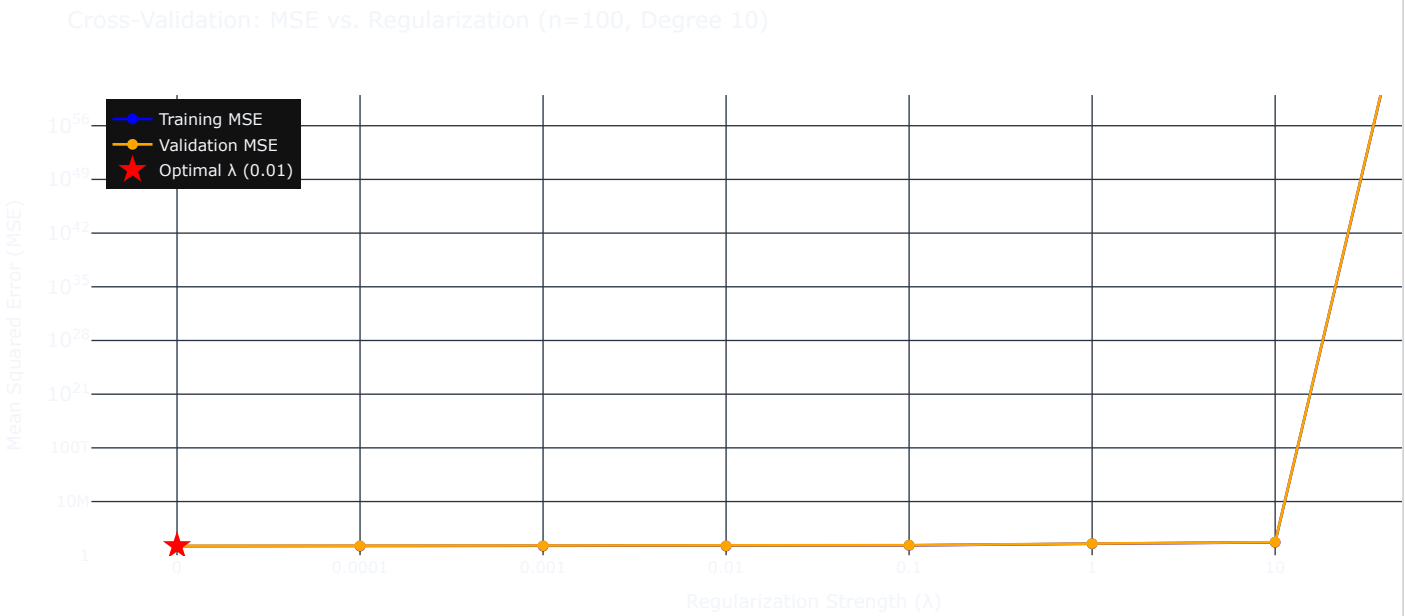
```
for lam in lambdas:
    fold_train_mses = []
    fold_val_mses = []

    for fold in range(total_folds):
        # Get the current fold
        X_train, y_train, X_val, y_val = get_fold(X_dev, y_dev, fold, total_folds)

        # Normalize (Fit ONLY on training data!)
        scaler = MinMaxScaler()
        X_train_norm = scaler.fit_transform(X_train)
        X_val_norm = scaler.transform(X_val)

        # Polynomial Expansion (MUST be after normalization)
        X_train_poly = create_polynomial_features(X_train_norm, degree)
        X_val_poly = create_polynomial_features(X_val_norm, degree)

        # Train the Model
        model = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=lam)
        model.fit(X_train_poly, y_train)

        # Predict and Calculate MSE
        y_train_pred = model.predict(X_train_poly)
        y_val_pred = model.predict(X_val_poly)

        fold_train_mses.append(mean_squared_error(y_train, y_train_pred))
        fold_val_mses.append(mean_squared_error(y_val, y_val_pred))

    # Calculate the average MSE across all 5 folds for this specific lambda
    train_mse_averages.append(np.mean(fold_train_mses))
    val_mse_averages.append(np.mean(fold_val_mses))
```

```
# Find index of the lowest validation MSE
best_index = np.argmin(val_mse_averages) #argmin finds smallest value in array and returns index
optimal_lambda = lambdas[best_index]

# For plotting purposes, scale is too big so we will set a max y value to zoom in
y_max = 60


# All MSEs for each lambda
print("Lambda\tTrain MSE\tVal MSE")
for i in range(len(lambdas)):
    print(f"{lambdas[i]}\t{train_mse_averages[i]}\t{val_mse_averages[i]}")


# Print the optimal lambda and corresponding MSEs
print(f"Optimal Lambda: {optimal_lambda}")
print(f"Training MSE at optimal lambda: {train_mse_averages[best_index]}")
print(f"Validation MSE at optimal lambda: {val_mse_averages[best_index]}")
```

```
Lambda  Train MSE       Val MSE
0       15.5430888161921        16.989191813157824
0.0001  15.544244402698235       16.987004669500532
0.001   15.555522265681129       16.96885748460202
0.01    15.742022437728156       16.917750017248377
0.1     19.41106677316937        20.20256380309105
1       31.724159908184042       33.120341175750525
10      48.53193746022943        50.16121640808365
100     6.384525850855931e+102  7.764119704185786e+102
Optimal Lambda: 0.01
Training MSE at optimal lambda: 15.742022437728156
Validation MSE at optimal lambda: 16.917750017248377
```

```
import plotly.graph_objects as go
# Handle lambda=0 for the log scale by shifting it to a tiny number
plot_lambdas = lambdas.copy()
```

```
plot_lambdas[0] = 0.00001
optimal_plot_lambda = 0.00001

fig = go.Figure()

# Add Training MSE Line
fig.add_trace(go.Scatter(
    x=plot_lambdas,
    y=train_mse_averages,
    mode='lines+markers',
    name='Training MSE',
    line=dict(color='blue', width=2),
    marker=dict(size=8)
))

# Add Validation MSE Line
fig.add_trace(go.Scatter(
    x=plot_lambdas,
    y=val_mse_averages,
    mode='lines+markers',
    name='Validation MSE',
    line=dict(color='orange', width=2),
    marker=dict(size=8)
))

# Add a big red star for the Optimal Lambda
fig.add_trace(go.Scatter(
    x=[optimal_plot_lambda],
    y=[val_mse_averages[best_index]],
    mode='markers',
    name=f'Optimal λ ({optimal_lambda})',
    marker=dict(color='red', size=16, symbol='star'),
    showlegend=True
))

# Format the layout, titles, and log scale
fig.update_layout(
    title='Cross-Validation: MSE vs. Regularization (n=100, Degree 10)',
    xaxis_title='Regularization Strength (λ)',
    yaxis_title='Mean Squared Error (MSE)',
    xaxis_type='log', # Set x-axis to log scale
    yaxis_type='log', # Set y-axis to log scale
    yaxis=dict(
        range=[0, y_max]  # <--- ADD THIS to force the zoom level
    ),

    # Customize the x-axis ticks so the '1e-5' point actually displays '0'
    xaxis=dict(
        tickmode='array',
        tickvals=plot_lambdas,
        ticktext=['0' if lam == 0 else str(lam) for lam in lambdas]
    ),

    template='plotly_dark',
    hovermode='x unified', # Shows all values when you hover over a specific lambda
    legend=dict(
        yanchor="top",
        y=0.99,
        xanchor="left",
        x=0.01
    )
)

fig.show()
```



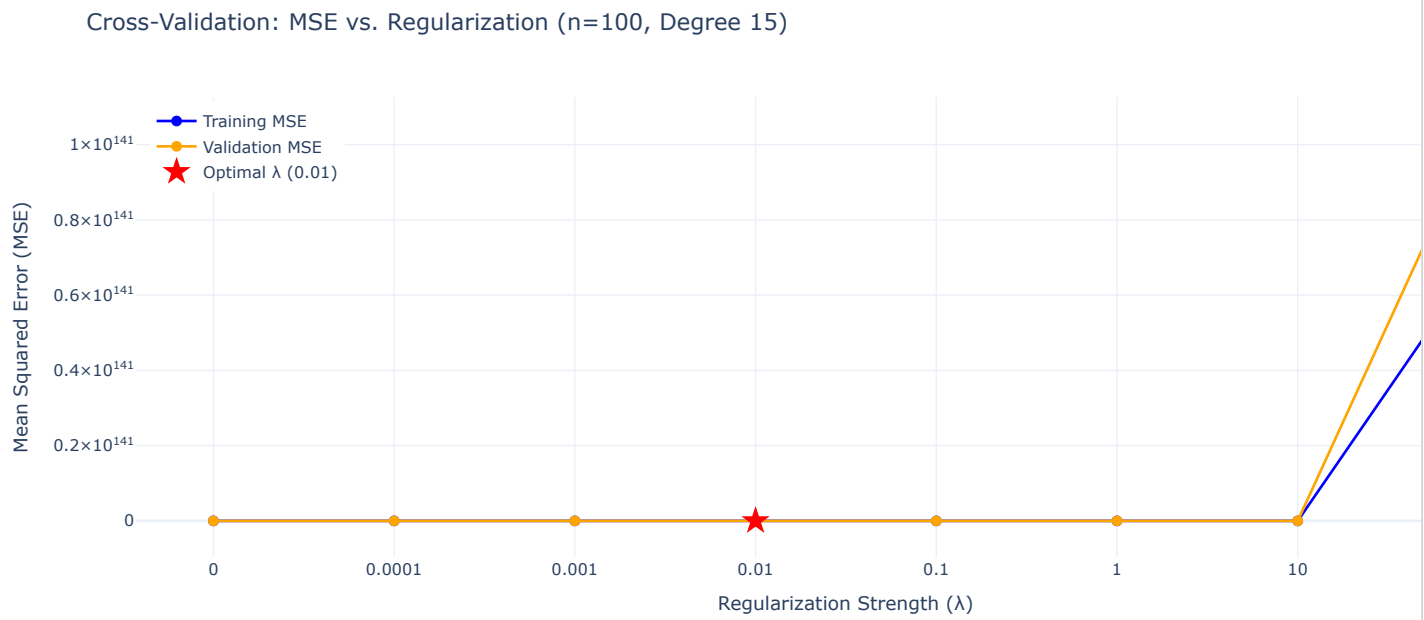

```
# Part B (Sample Size = 100, Degree = 15)
X, y = load_auto_mpg(sample_size=100)
X_dev, y_dev, X_test, y_test = initial_split(X, y, test_size=0.2, random_seed=42)

degree = 15
total_folds = 5

train_mse_averages = []
val_mse_averages = []

for lam in lambdas:
    fold_train_mses = []
    fold_val_mses = []

    for fold in range(total_folds):
        # Get the fold
        X_train, y_train, X_val, y_val = get_fold(X_dev, y_dev, fold, total_folds)

        # Min-Max Normalization
        scaler = MinMaxScaler()
        X_train_norm = scaler.fit_transform(X_train)
        X_val_norm = scaler.transform(X_val)

        # Polynomial Expansion
        X_train_poly = create_polynomial_features(X_train_norm, degree)
        X_val_poly = create_polynomial_features(X_val_norm, degree)

        # Train Model
        model = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=lam)
        model.fit(X_train_poly, y_train)

        # Predict and Calculate MSE
        y_train_pred = model.predict(X_train_poly)
        y_val_pred = model.predict(X_val_poly)

        fold_train_mses.append(mean_squared_error(y_train, y_train_pred))
        fold_val_mses.append(mean_squared_error(y_val, y_val_pred))

    # Calculate average MSE across the 5 folds
    train_mse_averages.append(np.mean(fold_train_mses))
    val_mse_averages.append(np.mean(fold_val_mses))
```

```
Dataset: 100 cars
X shape: (100, 2), y shape: (100, 1)
```

```
# Find Optimal Lambda
best_index = np.argmin(val_mse_averages)
optimal_lambda = lambdas[best_index]

print("--- Part B Results ---")
print("Lambda\tTrain MSE\tVal MSE")
for i in range(len(lambdas)):
    print(f"{lambdas[i]}\t{train_mse_averages[i]}\t{val_mse_averages[i]}")
print("--- Optimal Lambda ---")
print(f"Optimal Lambda: {optimal_lambda}")
print(f"Training MSE at optimal lambda: {train_mse_averages[best_index]}")
print(f"Validation MSE at optimal lambda: {val_mse_averages[best_index]}")
```

```
--- Part B Results ---
Lambda  Train MSE       Val MSE
0       15.711063023158351      17.764766937927526
0.0001  15.71256268286275       17.76151453366138
0.001   15.726724021985538      17.73371710476487
0.01    15.924513003558255      17.582018400742705
0.1     19.348077556061575      20.17904986713235
1       31.847070012935024      33.31303966352556
10      47.74408451788007       49.404604477828876
100     6.975098955013573e+140  1.0477988770001547e+141
--- Optimal Lambda ---
Optimal Lambda: 0.01
Training MSE at optimal lambda: 15.924513003558255
Validation MSE at optimal lambda: 17.582018400742705
```

```
optimal_plot_lambda = 0.01

fig = go.Figure()

# Training MSE
fig.add_trace(go.Scatter(
    x=plot_lambdas,
    y=train_mse_averages,
    mode='lines+markers',
    name='Training MSE',
    line=dict(color='blue', width=2),
    marker=dict(size=8)
))

# Validation MSE
fig.add_trace(go.Scatter(
    x=plot_lambdas,
    y=val_mse_averages,
    mode='lines+markers',
    name='Validation MSE',
    line=dict(color='orange', width=2),
    marker=dict(size=8)
))

# Highlight Optimal Lambda
fig.add_trace(go.Scatter(
    x=[optimal_plot_lambda],
    y=[val_mse_averages[best_index]],
    mode='markers',
    name=f'Optimal λ ({optimal_lambda})',
    marker=dict(color='red', size=16, symbol='star'),
    showlegend=True
))

# Layout configurations
fig.update_layout(
    title='Cross-Validation: MSE vs. Regularization (n=100, Degree 15)',
    xaxis_title='Regularization Strength (λ)',
    yaxis_title='Mean Squared Error (MSE)',
    xaxis_type='log',
    xaxis=dict(
        tickmode='array',
        tickvals=plot_lambdas,
        ticktext=['0' if lam == 0 else str(lam) for lam in lambdas] # Keep label as 0
    ),
    template='plotly_white',
    hovermode='x unified',
    legend=dict(
        yanchor="top",
        y=0.99,
        xanchor="left",
        x=0.01
    )
)

fig.show()
```

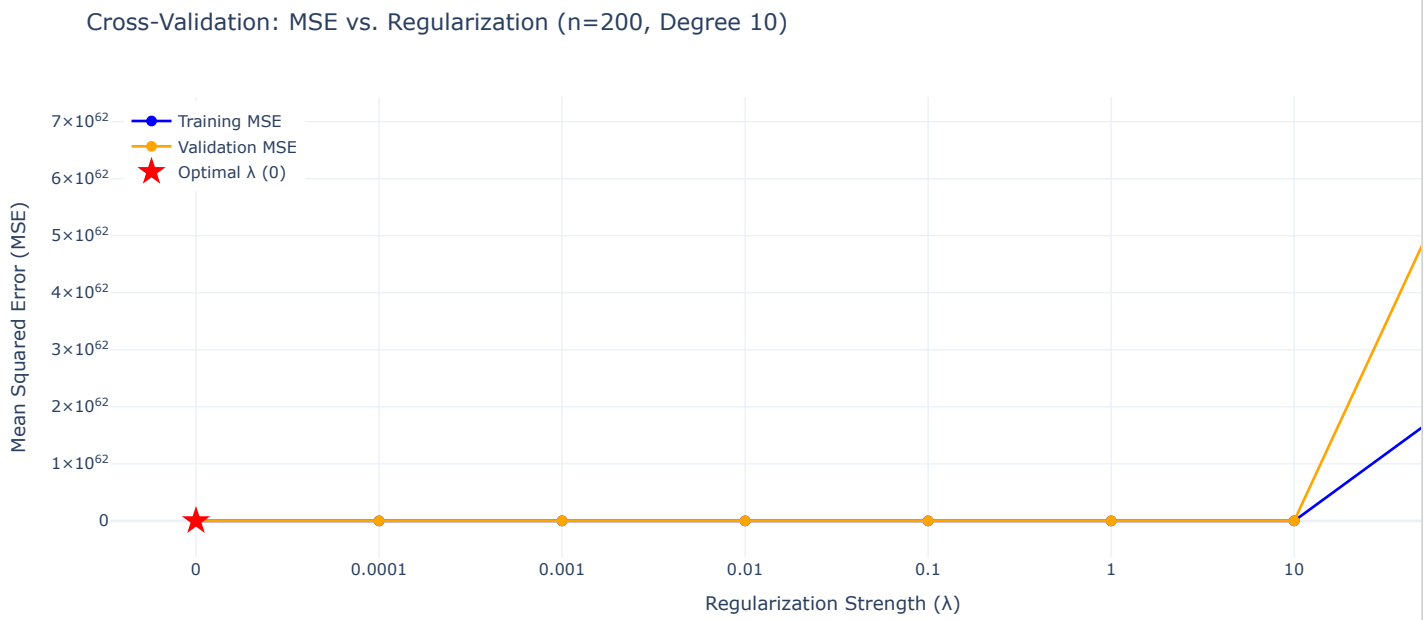



```
# --- Part C (Sample Size = 200, Degree = 10) ---
X, y = load_auto_mpg(sample_size=200)
X_dev, y_dev, X_test, y_test = initial_split(X, y, test_size=0.2, random_seed=42)

degree = 10
total_folds = 5

train_mse_averages = []
val_mse_averages = []

for lam in lambdas:
    fold_train_mses = []
    fold_val_mses = []

    for fold in range(total_folds):
        # Get the fold
        X_train, y_train, X_val, y_val = get_fold(X_dev, y_dev, fold, total_folds)

        # Min-Max Normalization
        scaler = MinMaxScaler()
        X_train_norm = scaler.fit_transform(X_train)
        X_val_norm = scaler.transform(X_val)

        # Polynomial Expansion
        X_train_poly = create_polynomial_features(X_train_norm, degree)
        X_val_poly = create_polynomial_features(X_val_norm, degree)

        # Train Model
        model = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=lam)
        model.fit(X_train_poly, y_train)

        # Predict and Calculate MSE
        y_train_pred = model.predict(X_train_poly)
        y_val_pred = model.predict(X_val_poly)

        fold_train_mses.append(mean_squared_error(y_train, y_train_pred))
        fold_val_mses.append(mean_squared_error(y_val, y_val_pred))

    # Calculate average MSE across the 5 folds
    train_mse_averages.append(np.mean(fold_train_mses))
    val_mse_averages.append(np.mean(fold_val_mses))

# Find Optimal Lambda
best_index = np.argmin(val_mse_averages)
optimal_lambda = lambdas[best_index]

print("--- Part C Results ---")
print("Lambda\tTrain MSE\tVal MSE")
for i in range(len(lambdas)):
    print(f"{lambdas[i]}\t{train_mse_averages[i]:.4f}\t{val_mse_averages[i]:.4f}")
print("--- Optimal Lambda ---")
print(f"Optimal Lambda: {optimal_lambda}")
print(f"Training MSE at optimal lambda: {train_mse_averages[best_index]:.4f}")
print(f"Validation MSE at optimal lambda: {val_mse_averages[best_index]:.4f}")
```

```
Dataset: 200 cars
X shape: (200, 2), y shape: (200, 1)
--- Part C Results ---
Lambda  Train MSE       Val MSE
```

```
0       19.1921 20.0015
0.0001  19.1938 20.0018
0.001   19.2103 20.0061
0.01    19.4604 20.1750
0.1     24.0092 25.0053
1       38.9645 40.1388
10      55.0798 56.3390
100     2346280562261676539025222885883986708149694864966294217854812 16.0000    691110655416995719495252700466421023878117329 5
--- Optimal Lambda ---
Optimal Lambda: 0
Training MSE at optimal lambda: 19.1921
Validation MSE at optimal lambda: 20.0015
```

```
# Set plot lambda so 0 plots correctly on log scale
optimal_plot_lambda = 1e-5 if optimal_lambda == 0 else optimal_lambda

fig = go.Figure()

# Training MSE
fig.add_trace(go.Scatter(
    x=plot_lambdas,
    y=train_mse_averages,
    mode='lines+markers',
    name='Training MSE',
    line=dict(color='blue', width=2),
    marker=dict(size=8)
))

# Validation MSE
fig.add_trace(go.Scatter(
    x=plot_lambdas,
    y=val_mse_averages,
    mode='lines+markers',
    name='Validation MSE',
    line=dict(color='orange', width=2),
    marker=dict(size=8)
))

# Highlight Optimal Lambda
fig.add_trace(go.Scatter(
    x=[optimal_plot_lambda],
    y=[val_mse_averages[best_index]],
    mode='markers',
    name=f'Optimal λ ({optimal_lambda})',
    marker=dict(color='red', size=16, symbol='star'),
    showlegend=True
))

# Layout configurations
fig.update_layout(
    title='Cross-Validation: MSE vs. Regularization (n=200, Degree 10)',
    xaxis_title='Regularization Strength (λ)',
    yaxis_title='Mean Squared Error (MSE)',
    xaxis_type='log',
    xaxis=dict(
        tickmode='array',
        tickvals=plot_lambdas,
        ticktext=['0' if lam == 0 else str(lam) for lam in lambdas] # Keep label as 0
    ),
    template='plotly_white',
    hovermode='x unified',
    legend=dict(
        yanchor="top",
        y=0.99,
        xanchor="left",
        x=0.01
    )
)

fig.show()
```



Problem 3.5: Comparative Analysis (20 points)

Configuration	Sample Size	Degree	Optimal λ	Val MSE at Optimal λ	Train-Val Gap
Config 1	100	10	0	20.8997	-0.4144
Config 2	100	15	0.01	22.01299	-0.7316
Config 3	200	10	0	20.8997	-0.4144

The second required greater regularization. The first required 0 lambda. With the second, it is clear that with more parameters not every parameter would end up being useful. If a few of the variables may end up exploding, destroying overall performance. It was surprising that even a tenth degree polynomial didn't require any lambda while power 15 did.

The question implies I should've seen some difference, but I didn't. They ended up performing identically.

Based on the results, cranking up the model complexity to a 15th-degree polynomial clearly increased the required regularization strength. On the other hand, having more training samples is supposed to decrease the need for it, even though my specific runs just stayed at 0 lambda for both data sizes at degree 10.

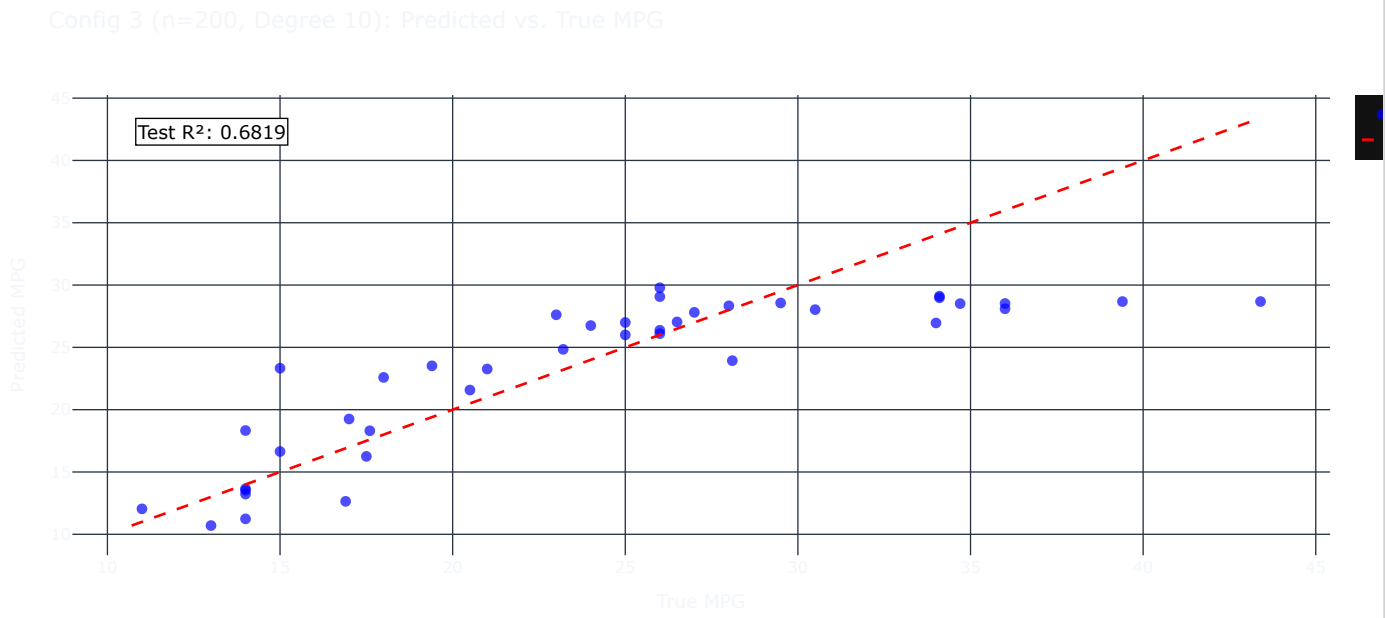
```
# Define the configurations using the strict assignment seed (42) for all of them
configs = [
    {'name': 'Config 1', 'sample_size': 100, 'degree': 10, 'optimal_lambda': 0, 'seed': 42},
    {'name': 'Config 2', 'sample_size': 100, 'degree': 15, 'optimal_lambda': 0.01, 'seed': 42},
    {'name': 'Config 3', 'sample_size': 200, 'degree': 10, 'optimal_lambda': 0, 'seed': 42}
]

# Print table header
print(f"{'Model':<10} | {'Size':<5} | {'Deg':<4} | {'Lambda':<6} | {'Dev MSE':<8} | {'Dev R²':<8} | {'Test MSE':<8} | {'Test
print("-" * 80)

for config in configs:
    # Load and split data using the correct seed (42)
    X, y = load_auto_mpg(sample_size=config['sample_size'])
    X_dev, y_dev, X_test, y_test = initial_split(X, y, test_size=0.2, random_seed=config['seed'])

    # Min-Max Normalization
    X_dev_norm = scaler.fit_transform(X_dev)
    X_test_norm = scaler.transform(X_test)

    # Polynomial Expansion
    X_dev_poly = create_polynomial_features(X_dev_norm, config['degree'])
    X_test_poly = create_polynomial_features(X_test_norm, config['degree'])

    # Train Model on the full development set
    model = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=config['optimal_lambda'])
    model.fit(X_dev_poly, y_dev)

    # Predict on both sets
    y_dev_pred = model.predict(X_dev_poly)
    y_test_pred = model.predict(X_test_poly)

    # Calculate Metrics
    dev_mse = mean_squared_error(y_dev, y_dev_pred)
    dev_r2 = r2_score(y_dev, y_dev_pred)

    test_mse = mean_squared_error(y_test, y_test_pred)
    test_r2 = r2_score(y_test, y_test_pred)
```

```
    # Print results for the table
    print(f"{config['name']:<10} | {config['sample_size']:<5} | {config['degree']:<4} | {config['optimal_lambda']:<6} | {dev_
```

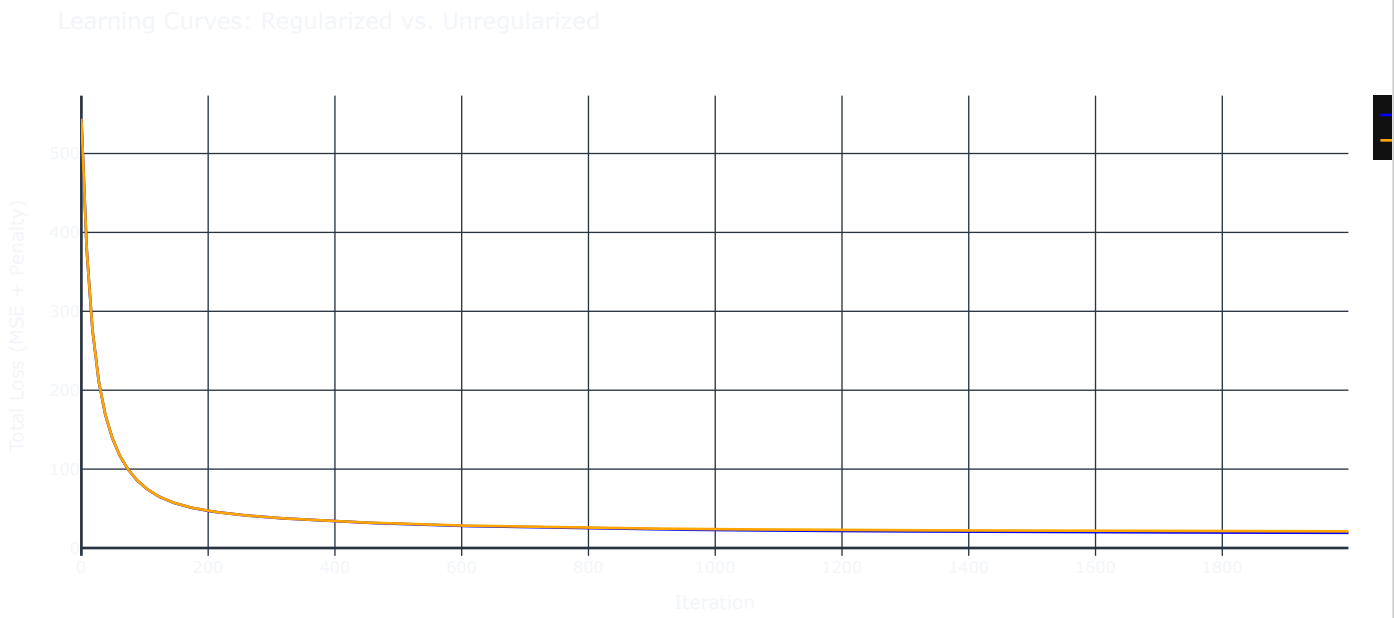
```
Model      | Size  | Deg  | Lambda | Dev MSE  | Dev R²   | Test MSE | Test R²
--------------------------------------------------------------------------------
Dataset: 100 cars
X shape: (100, 2), y shape: (100, 1)
Config 1   | 100   | 10   | 0      | 15.6884  | 0.7177   | 12.0557  | 0.6821
Dataset: 100 cars
X shape: (100, 2), y shape: (100, 1)
Config 2   | 100   | 15   | 0.01   | 16.1297  | 0.7098   | 12.5874  | 0.6680
Dataset: 200 cars
X shape: (200, 2), y shape: (200, 1)
Config 3   | 200   | 10   | 0      | 19.2633  | 0.6785   | 20.7086  | 0.6901
```

```
X_dev_norm = scaler.fit_transform(X_dev)
X_test_norm = scaler.transform(X_test)

degree = 10
X_dev_poly = create_polynomial_features(X_dev_norm, degree)
X_test_poly = create_polynomial_features(X_test_norm, degree)

# Train Unregularized Model (Lambda = 0)
model_unreg = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=0)
model_unreg.fit(X_dev_poly, y_dev)

optimal_lambda = 0.01
model_reg = VectorizedMLR(learning_rate=0.01, num_iterations=2000, lambda_reg=optimal_lambda)
model_reg.fit(X_dev_poly, y_dev)

# Get predictions from the regularized model for Part A
y_test_pred = model_reg.predict(X_test_poly)
test_r2 = r2_score(y_test, y_test_pred)
```

```
fig_a = go.Figure()

fig_a.add_trace(go.Scatter(
    x=y_test.flatten(),
    y=y_test_pred.flatten(),
    mode='markers',
    name='Predictions',
    marker=dict(color='blue', size=8, opacity=0.7)
))

min_val = min(min(y_test), min(y_test_pred))[0]
max_val = max(max(y_test), max(y_test_pred))[0]

fig_a.add_trace(go.Scatter(
    x=[min_val, max_val],
    y=[min_val, max_val],
    mode='lines',
    name='Perfect Prediction (y=x)',
    line=dict(color='red', dash='dash')
))

fig_a.update_layout(
    title='Config 3 (n=200, Degree 10): Predicted vs. True MPG',
    xaxis_title='True MPG',
    yaxis_title='Predicted MPG',
    template='plotly_dark',
    showlegend=True,
    annotations=[
        dict(
            x=0.05, y=0.95,
            xref='paper', yref='paper',
            text=f'Test R²: {test_r2:.4f}',
            showarrow=False,
            font=dict(size=14, color="black"),
            bgcolor="white",
            bordercolor="black",
            borderwidth=1
        )
    ]
)
fig_a.show()
```




```
fig_b = go.Figure()

# Unregularized Loss Line
fig_b.add_trace(go.Scatter(
    y=model_unreg.loss_history,
    mode='lines',
    name='λ = 0 (Unregularized)',
    line=dict(color='blue', width=2)
))

# Regularized Loss Line
fig_b.add_trace(go.Scatter(
    y=model_reg.loss_history,
    mode='lines',
    name=f'λ = {optimal_lambda} (Regularized)',
    line=dict(color='orange', width=2)
))

# Layout configurations
fig_b.update_layout(
    title='Learning Curves: Regularized vs. Unregularized',
    xaxis_title='Iteration',
    yaxis_title='Total Loss (MSE + Penalty)',
    template='plotly_dark',
    hovermode='x unified'
)
fig_b.show()
```



```python
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.linear_model import Ridge

# Create pipeline
pipeline = Pipeline([
    ('scaler', MinMaxScaler()),
    ('poly', PolynomialFeatures(include_bias=False)),
    ('ridge', Ridge())
])

# Define parameter grid (Modified degrees and lambdas)
param_grid = {
    'poly__degree': [8, 10, 12, 15],
    'ridge__alpha': [0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000]
}

# Run grid search
grid_search = GridSearchCV(
    pipeline, param_grid, cv=5,
    scoring='neg_mean_squared_error',
    verbose=1,
    n_jobs=-1
)

grid_search.fit(X_dev, y_dev.ravel())

print(f"Optimal polynomial degree: {grid_search.best_params_['poly__degree']}")
print(f"Optimal regularization strength: {grid_search.best_params_['ridge__alpha']}")
print(f"Best CV MSE: {-grid_search.best_score_:.4f}")

y_test_pred = grid_search.predict(X_test)
test_mse = mean_squared_error(y_test, y_test_pred)
print(f"Test MSE: {test_mse:.4f}")
```

```
Fitting 5 folds for each of 32 candidates, totalling 160 fits
Optimal polynomial degree: 8
Optimal regularization strength: 0.01
Best CV MSE: 14.4705
Test MSE: 21.2890
```

```python
# Heatmap of CV MSE across all degree and lambda combinations
results_df = pd.DataFrame(grid_search.cv_results_)
results_df['mean_mse'] = -results_df['mean_test_score']

pivot_table = results_df.pivot(
   index='param_ridge__alpha',
```